

Birla Institute of Technology and Sciences Pilani,
KK Birla Goa Campus
Machine Learning for Electronics Engineers
EEE G513
Comprehensive Examination (Part A)

Max Time: 1Hr

Max Marks: 35

Date: 4th December 2025

Mode: Open book, only paper based notes allowed. You can refer to slides on Quanta

Instructions:

1. In case of MCQs, 1 answer is correct. No negative marking.
2. In case of numerical questions, write the answer correct to 2 decimal places
3. Answer all questions. Partial marking will be done for subjective questions.

All the best!

Name:

ID Number:

(2 marks) Q1. Given d -dimensional data $\{x_i\}_{i=1}^N$, you run principle component analysis and pick P principle components. Can you always reconstruct any data point x_i for $i \in \{1 \dots N\}$ from the P principle components with zero reconstruction error?

A) No, some error will always be there no matter how many components you choose		C) Yes, if $P = d$	
B) Yes, if $P = d/2$		D) Yes, if $d/2 < P < d / \log(n)$	

(2 marks) Q2. What is true of k-means clustering?

A) k-means is a supervised learning algorithm		C) k-means clustering always converges to the same solution regardless of how clusters are initialized.	
B) Increasing k can never increase the optimal value of the k-means cost function.		D) The k-means algorithm that we learnt in class gives the globally optimum value of the k-means loss function	

Q3. Consider an arithmetic network with the inputs a , b , and c , which computes the following sequence of operations, where $s(\gamma) = \frac{1}{1 + e^{-\gamma}}$ is the logistic (sigmoid) function and $r(\gamma) = \max(0, \gamma)$ is the hinge function used by ReLUs.

$$d = ab, e = s(d), f = r(a), g = 3a, h = 2e + f + g, i = ch, j = f + i^2$$

(5 marks) 3.1 Construct the computational graph for the above expression. Represent variables (including intermediate ones) with **circles** and operators (like $+$, $*$, s etc) with a **square**.

(5 marks) 3.2 We want to find the partial derivatives of j with respect to every other variable a through i , in backpropagation style. This means that for each variable z , we want you to write $\frac{\partial j}{\partial z}$ in two forms: (1) in terms of derivatives involving each variable that directly uses the value of z , and (2) in terms of the inputs and intermediate values $a \cdots i$, as simply as possible but with no derivative symbols. For example, we write

$$\frac{\partial j}{\partial i} = \frac{dj}{di} = 2i \text{ (no chain rule needed)}$$

$$\frac{\partial j}{\partial h} = \frac{\partial j}{\partial i} \frac{\partial i}{\partial h} = 2ic \text{ (chain rule, then backprop the derivative expressions)}$$

Q. Find $\frac{\partial j}{\partial b}$. If they are needed, express the derivative $s'(\gamma)$ in terms of $s(\gamma)$ and express the derivative $r'(\gamma)$ as the indicator function $\mathbf{1}(\gamma \geq 0)$.

(2 marks) Q4. Which of the following are benefits of using convolutional neural networks—as opposed to fully connected ones— for image recognition tasks?

A: The ability to express a wider variety of more complicated functions of the input features

B: Fewer model architecture hyperparameters for the designer to select

C: Enables the network to more easily learn and recognize features regardless of their position in the image

D: Typically requires less data to train well

A) All are correct		C) A, C	
B) A, B, D		D) C, D	

(3 marks) Q5. Select the true statements about regularization in regression.

A: Adding ℓ_2 regularization to a regression problem never decreases the training cost (mean loss over the training points) at the global minimum

B: Adding ℓ_2 regularization to a regression problem never decreases the test cost (mean loss over the test points) at the training points' global minimum.

C: ℓ_1 regularization is a good method for feature selection.

D: ℓ_2 regularization is a good method for feature selection.

A) A, C		C) A, B	
B) A, B, C		D) C, D	

(3 marks) Q6. In a convolutional neural network, which of the following changes to network parameters will decrease the size (number of units) of a hidden layer of the network?

A: Increasing the size of the filters (masks) in a convolutional layer. (Assume we do not use any padding nor "periodic boundaries.")

B: Increasing the number of filters (masks) in a convolutional layer.

C: Changing a pooling layer to use a 4×4 sliding window with stride 4 instead of a 2×2 sliding window with stride 2.

D: Decreasing the size of the mini-batch used for stochastic gradient descent.

A) All are correct		C) A, C	
B) A, B, D		D) C, D	

(3 marks) Q7. You are training a neural network, but the training error is high. Which of the following, if done in isolation, has a better-than-tiny chance of reducing the training error?

A: Adding another hidden layer

B: Normalizing the input data

C: Adding more units to hidden layers

D: Training on more data

A) All are correct		C) A, C	
B) None are correct		D) A, B, C	

(10 marks) Q8. You are given a dataset of 10 points in a 2D plane and a requirement to cluster them into **K=3** groups. You must perform the K-Means algorithm manually for **one full iteration** (Assignment Step + Update Step).

Point ID - (X,Y)

P1 - (2,2)

P2 - (3,2)

P3 - (2,3)

P4 - (10,2)

P5 - (11,2)

P6 - (10,3)

P7 - (6,8)

P8 - (7,8)

P9 - (6,9)

P10 - (7,9)

Parameters:

1. **Number of Clusters (K):** 3
2. **Distance Metric:** Manhattan Distance (L_1 Norm).
 - Formula: $D = |x_2 - x_1| + |y_2 - y_1|$
3. **Initial Centroids:**
 - $C_1 : (2,2); C_2 : (10,2); C_3 : (6,8)$

